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Aryabhatta Knowledge University, Patna

B.Tech 5th Semester Exam., 2021 (New Course)

Subject: Hydrology and Water Resources Engineering | ***Time: 03 Hours*** | ***Code: 101505*** | ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) An unconfined aquifer is one in which
(i) water surface under the ground is at atmospheric pressure
(ii) water is under pressure greater than the atmospheric pressure between impervious strata
(iii) water is at pressure below the atmospheric pressure between impervious strata
(iv) water is either below or above the atmospheric pressure between impervious strata

(b) A unit hydrograph has one unit of
(i) direct runoff
(ii) rainfall duration
(iii) peak discharge
(iv) the time base of direct runoff

(c) The instrument to measure wind velocity is
(i) current meter
(ii) atmometer
(iii) aerometer
(iv) anemometer

(d) For a basin, in a given period Δt, there is no change in the groundwater and soil water status. If P = precipitation, R = total runoff, E = Evapotranspiration and ΔS = increase in the surface water storage basin, the hydrological water budget equation states
(i) $P = R - E \pm \Delta S$
(ii) $R = P + E - \Delta S$
(iii) $P = R + E + \Delta S$
(iv) None of the above

(e) The instrument to measure humidity variation in atmosphere is
(i) barograph
(ii) thermograph
(iii) hygrograph
(iv) thermo-hygrograph

(f) For a given storm, other factors remaining same
(i) basins having low drainage density give smaller peaks in flood hydrographs
(ii) basins with larger drainage densities give smaller flood peaks
(iii) low drainage density basins give shorter time bases of hydrographs
(iv) the flood peak is independent of the drainage density

(g) Most accurate method of finding average depth of rainfall over an area is
(i) isohyetal method
(ii) arithmetic mean method
(iii) Thiessen polygon method
(iv) mass curve method

(h) Base-flow separation is performed
(i) on a unit hydrograph to get the direct-runoff hydrograph
(ii) on a flood hydrograph to obtain the magnitude of effective rainfall
(iii) on flood hydrographs to obtain the rainfall hyetograph
(iv) on hydrographs of effluent streams only

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(i) Virgin flow means
   (i) the flow in the stream corresponding to no infiltration and evaporation losses
   (ii) the flow in the stream which is not affected by the works of human
   (iii) the flow in the stream which does not contain flow from watershed leakage of the neighboring basin
   (iv) the flow in the stream corresponding to no base-flow contribution

(j) Flow duration curve is the graph between
   (i) discharge in the stream and time
   (ii) accumulated discharge in the stream and time
   (iii) discharge and percent of time such discharge is equalled or exceeded
   (iv) discharge and gauge height

Q.2 Solve this question: **[14]**

A stream size of 150 lit/sec was released from the diversion headwork to irrigate a land of area 18 hectare. The stream size when measured at the delivery to the field channels is 120 lit/sec. The stream was continued for 8 hours. The effective root zone depth is 1.80 m. The application losses in the field are estimated to be 440 m^3. The depth of water penetration was 1.80 m and 1.20 m at the head and tail of the run respectively. The available water holding capacity of the soil is 21 cm/m and irrigation was done at 60% depletion of available moisture (A_m). Find conveyance efficiency (E_c), water application efficiency (E_a), water storage efficiency (E_s) and water distribution efficiency (E_d). The stream size delivered to the plot was 100 lit/sec.

Q.3 Solve this question: **[14]**

An irrigation channel is to be designed for a discharge of 50 cumecs adopting the available ground slope of 1.5×10^{-4}. The riverbed material has a median size of 2.0 mm. Design the channel and recommend the size of the coarser material to be excluded or ejected from the channel for its efficient functioning. Use Lacey theory. Assume side slope of 0.5 H: 1 V. The following are the rates of rainfall for successive 30 minutes period for storm duration of 210 minutes: 5.5; 6.0; 12.5; 8.0; 3.25; 3.25; 6.5 cm/hr. Take φ-index as 4.5 cm/hr. Calculate the total runoff (in cm) and total rainfall.

Q.4 Solve both questions: **[14]**

(a) Discuss SCS-CN method of estimating runoff volume.

(b) Land use and corresponding runoff coefficient are as given below:

Land use	Area (ha)	Runoff coefficient
Roads	8	0.70
Lawn	17	0.10
Residential area	50	0.30
Industrial area	10	0.80

Calculate the equivalent runoff coefficient.

Q.5 Solve this question: **[14]**

A 30 cm well completely penetrates an unconfined aquifer of saturated depth 40 m. After a long period of pumping at a steady rate of 1500 lpm, the drawdown, in two observation wells 25 m and 75 m from the pumping well were found to be 3.5 m and 2.0 m respectively. Determine the transmissivity of the aquifer. What is the drawdown at the pumping well?

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Q.6 Solve this question: **[14]**

Determine the location of closed tile drains below ground for the following data:
Root zone depth = 1.5 m
Capillary rise in soil = 0.3 m
Coefficient of permeability of soil = 1.5×10^{-4} m/s
Drainage capacity = 0.11 $m^3/s/km^2$
Spacing of drains = 200 m
Depth of impervious stratum below ground = 10.0 m

Q.7 Solve this question: **[14]**

Discuss elementary profile of gravity. How are its dimensions fixed? Also derive normal stresses and principal stresses on the elementary profile.

Q.8 Solve this question: **[14]**

Use the information from the table given below to find moisture content in the root zone at different depths, depth of water available in the root zone at different depths, total depth of water available in the root zone and the soil moisture deficit. Assuming peak rate of consumptive use as 8 mm/day also find the irrigation interval:

Depth of sampling (m)	Wt. of moist soil sample (gm)	Oven dry wt. (gm)
0.0 - 0.30	98.80	94.60
0.30 - 0.60	96.60	92.10
0.60 - 0.90	95.00	90.60
0.90 - 1.20	94.00	89.40

The bulk density of the soil in the root zone was 1.60 gm/cc. The moisture holding capacity of the soil at field capacity was 19.60 cm/m depth.

Q.9 Solve this question: **[14]**

The GCA for a distributary is 5000 hectare (ha) 80% of which is CCA. The intensity of irrigation for Rabi is 50% and for Kharif is 30%. The average duty at the head of the distributary is 2000 ha/cumec for Rabi and 900 ha/cumec for Kharif. Determine discharge at the head of distributary. What will be design discharge if conveyance efficiency is 70%, time factor is 0.8 and capacity factor is 0.9?

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